



CIMA Management Case Study

Mock Exam 4

Flatthall

Unseen

Task 1 - Unseen case material [45 minutes]

The following email arrives in your inbox.

From: Susanna Leong, Senior Financial Manager

To: Financial Manager

Subject: New FlattKey app

Hi Financial Manager,

The Board is currently considering a proposal to introduce a new app - FlattKey - for students living in Flatthall accommodation. I have attached a copy of this proposal.

Please could you write a report in which you:

- Analyse the potential impact on stakeholders and discuss how we should effectively communicate with them if the FlattKey app proposal goes ahead.

[Sub task a - 60%]

- Discuss how the project to develop and implement the FlattKey app should be organised and managed.

[Sub task b - 40%]

Kind regards,
Susanna

Reference material

Proposal: FlattKey: A Digital Key & Maintenance Hub

Executive Summary

The proposed FlattKey app is a comprehensive mobile application designed to serve as a digital key for students' accommodations and a centralised platform for reporting maintenance issues within Flatthall properties. This innovation aligns with Flatthall's commitment to providing a high-quality living experience, leveraging technology to increase efficiency, enhance security and streamline communication between students and property management.

Features

Digital door key:

- Secure access - Utilises advanced encryption to ensure secure access to building entrances, individual flats and rooms.
- Lock replacement - Implements a property-wide replacement of traditional locks with advanced digital locks compatible with the app.

Maintenance reporting:

- Issue reporting - Enables residents to report maintenance issues directly within the app, including the option to upload photos and videos.
- Schedule repairs - Allows residents to select preferred times for maintenance work, minimising disruptions to their schedules.

Community engagement:

- Notifications - Sends alerts about upcoming maintenance.
- Feedback - Collects resident feedback on facilities and services to continuously improve the living experience.

Impact on concierge team

The introduction of the FlattKey app is designed to streamline operations by automating access control and maintenance reporting: tasks that are currently managed by the concierge team. As these processes become digital, there will no longer be the need for a concierge team, making these positions redundant.

Task 2 - Unseen case material [45 minutes]

It is nine months later. The following email arrives in your inbox.

From: Susanna Leong, Senior Financial Manager
To: Financial Manager
Subject: App security and utilising collected data

Hi Financial Manager,

As you may be aware, the FlattKey app was implemented for all of our locations over the summer holidays. It is now being used by students across the country to access accommodation and report maintenance issues, replacing our previous concierge service.

We have started to collect and store large amounts of data collected by the app. I have attached a document that outlines the types of data being collected.

Could you please write a report in which you:

- Suggest measures to ensure the ongoing reliability and security of the FlattKey app and its associated data.

[Sub task a - 50%]

- Advise on how the collected data could be leveraged to inform budgeting and investment decisions.

[Sub task b - 50%]

Kind regards,
Susanna

Reference material

FlattKey app - Types of collected data

User information:

- Student's full name and date of birth
- Student ID and institution details
- Email address and phone number
- Room number and building location
- Type of device, operating system and app version used to access the app

Access data:

- Dates, times and locations the digital key is used to access buildings and rooms

Maintenance data:

- Details of maintenance issues reported, including descriptions, photos or videos
- Dates and times selected by the user for carrying out repairs
- Resolution status and updates on reported issues

User engagement data:

- User-submitted feedback on accommodation facilities and maintenance services
- Frequency and duration of app use, most accessed features and navigation paths within the app

Task 3 - Unseen case material [45 minutes]

Another six months have passed. The following email arrives in your inbox.

From: Susanna Leong, Senior Financial Manager

To: Financial Manager

Subject: Proposal to introduce a user-pay model for utilities

Hi Financial Manager,

Following recent news highlighting the high energy wastage in halls of residence and purpose-built student accommodation, the Board is considering whether to implement smart tracking of utilities and transition to a user-pay model to promote energy conservation and sustainability. I have attached a copy of this proposal.

Please could you write a report in which you:

- Identify the key risks and uncertainties associated with introducing a user-pay model for utilities and suggest ways to mitigate these risks.

[Sub task a - 50%]

- Discuss the benefits and drawbacks of involving representatives from each department in the project's initial planning, as well as the importance of considering non-financial factors in assessing the success of this initiative.

[Sub task b - 50%]

Kind regards,
Susanna

Reference material

Proposal for implementing smart utility tracking and user-pay model at Flatthall properties

Introduction

In response to recent news highlighting the issue of high energy wastage in student accommodation where utilities are included in the rent, Flatthall is proposing the adoption of smart utility tracking technologies and a transition to a user-pay model. Traditionally, utility costs have been included in the rental agreements, offering simplicity for students but inadvertently encouraging less mindful consumption of resources. This initiative aims to promote energy conservation, reduce wastage and foster a culture of sustainability among students.

Objectives

- Encourage responsible consumption habits among residents
- Enable students who consume less to benefit from lower utility bills

Proposal

- 1. Installation of smart utility meters in private rooms** - Each student's private room will be equipped with smart meters to monitor their real-time energy use.
- 2. Smart water and energy tracking for showers** - Smart shower heads will be installed in all bathrooms, allowing water usage and energy consumption for heating water to be tracked. Students will require their digital FlattKey keycard to operate showers, linking water and energy usage to their personal profile.
- 3. Smart laundry facilities** - Laundry machines will be upgraded to smart models, allowing precise tracking of water and energy usage. Students will require their digital FlattKey keycard to operate laundry machines, linking water and energy usage to their personal profiles.
- 4. Integration with the FlattKey app** - Students will be able to view their real-time energy and water usage relating to their private room, showering and use of laundry facilities in the FlattKey app. They will receive a monthly bill for their usage, which can be paid via the app.

Task 4 - Unseen case material [45 minutes]

The following email arrives in your inbox.

From: Susanna Leong, Senior Financial Manager

To: Financial Manager

Subject: Social networking tools & maintenance delays

Hi Financial Manager,

There are a couple of matters I'd like you to help me with. Firstly, the Board is considering whether we should integrate social networking tools into the FlattKey app. These would include the introduction of an online forum for each location, event calendars and peer-to-peer support networks to foster a sense of belonging and connectivity within our student communities. Please could you write a report in which you:

- Evaluate the impact of incorporating social networking tools into the FlattKey app on Flatthall's business model.

[Sub task a - 40%]

Secondly, we've been receiving a large number of complaints about delayed responses to maintenance requests submitted through the app, and some concerns about how this issue has been managed have been raised. I've attached an extract of the latest Board minutes relating to this. In your report, could you also please:

- Evaluate the effectiveness of the leadership of the Operations Director and suggest an appropriate leadership style to ensure the situation is resolved effectively.

[Sub task b - 60%]

Kind regards,

Susanna

Reference material

Extract from Board minutes

Item 6: Review of operations post-digital initiatives rollout

Discussion

The Board reviewed the recent implementation of the FlattKey app and the replacement of the concierge teams across Flatthall properties. It was noted that there has been a significant uptick in complaints about delayed responses to maintenance requests submitted through the app. These delays are contributing to a growing dissatisfaction among students, impacting their overall experience and perception of Flatthall's management.

Operations Director's report

Dai Linlin, Operations Director, provided an update on the management of maintenance requests, specifically highlighting the delegation of responsibilities to mid-level managers at Flatthall's Head Office for the oversight of the independent contractors who undertake repair and maintenance.

Observations by Board members

- Several board members expressed concern about whether mid-level managers were adequately prepared for their new responsibilities. It was noted that there was no mention of additional training or resources being made available to these managers for managing independent contractors.
- Concerns were raised about the apparent lack of established communication channels for these managers to escalate issues or seek guidance, which could be contributing to the delays and inconsistencies in handling maintenance requests.

Present value tables

Present value of US\$1 i.e. $(1 + r)^{-n}$ where r = interest rate, n = number of periods until payment or receipt.

Period s (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.962	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.905	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.788	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149

Period s (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.901	.893	.885	.877	.870	.862	.855	.847	.840	.833
2	.812	.797	.783	.769	.756	.743	.731	.718	.706	.694
3	.731	.712	.693	.675	.658	.641	.624	.609	.593	.579
4	.659	.636	.613	.592	.572	.552	.534	.516	.499	.482
5	.593	.567	.543	.519	.497	.476	.456	.437	.419	.402
6	.535	.507	.480	.456	.432	.410	.390	.370	.352	.335
7	.482	.452	.425	.400	.376	.354	.333	.314	.296	.279
8	.434	.404	.376	.351	.327	.305	.285	.266	.249	.233
9	.391	.361	.333	.308	.284	.263	.243	.225	.209	.194
10	.352	.322	.295	.270	.247	.227	.208	.191	.176	.162
11	.317	.287	.261	.237	.215	.195	.178	.162	.148	.135
12	.286	.257	.231	.208	.187	.168	.152	.137	.124	.112
13	.258	.229	.204	.182	.163	.145	.130	.116	.104	.093
14	.232	.205	.181	.160	.141	.125	.111	.099	.088	.078
15	.209	.183	.160	.140	.123	.108	.095	.084	.074	.065
16	.188	.163	.141	.123	.107	.093	.081	.071	.062	.054
17	.170	.146	.125	.108	.093	.080	.069	.060	.052	.045
18	.153	.130	.111	.095	.081	.069	.059	.051	.044	.038
19	.138	.116	.098	.083	.070	.060	.051	.043	.037	.031
20	.124	.104	.087	.073	.061	.051	.043	.037	.031	.026

Cumulative present value tables

This table shows the Present Value of US\$1 per annum, Receivable or Payable at the end of each year for n years.

$$\frac{1 - (1+r)^{-n}}{r}$$

Periods (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	17.226	15.679	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	18.046	16.351	14.878	13.590	12.462	11.470	10.594	9.818	9.129	8.514
Periods (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870